

# Lecture 02

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## Review of Lecture 1

Covered

- Inductive vs deductive reasoning
- Formulating research questions
- Accuracy vs precision
- Data types and classifications
- Setting up R projects
- Installing and loading libraries
- Reading files into R
- Creating basic graphs

## Lecture 2: Project Design & Data Visualization

**The objectives:**

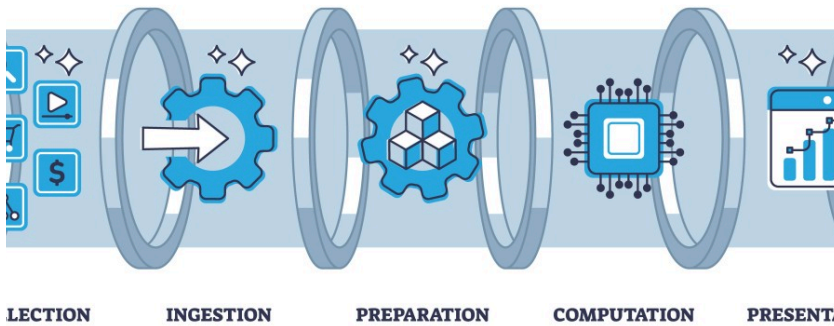
1. Design a well-organized project
2. Implement good naming conventions
  - Controlled vocabulary
  - Including units in names
3. Create and use metadata effectively
4. Build tidy, well-structured spreadsheets
5. Understand data repositories
6. Create effective visualizations with ggplot2



### Project Design: Step 1

- Data: the raw material of science
- Wide variety of formats, sizes, complexity
- Data management and curation often under emphasized
- Good data management: owe it to our funding agencies, colleagues, supervisors, and study systems

## DATA PIPELINE

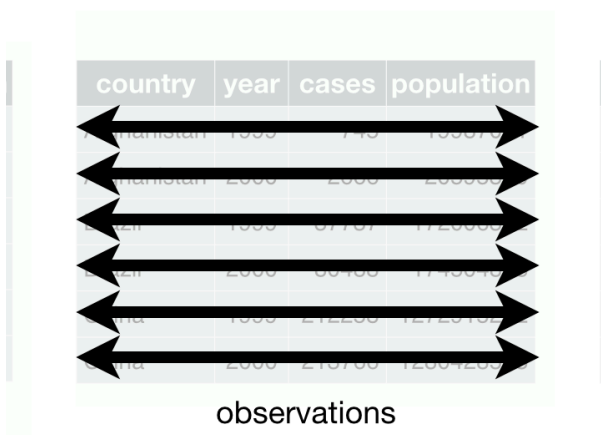


## Lecture 2: Project Design: Step 1

1. **Determine data types** you'll collect
2. **Establish controlled vocabulary**
  - Example: `do_mgl` for dissolved oxygen in mg/L
  - Example: `drp_uql` for dissolved reactive phosphorus in  $\mu\text{g/L}$
3. **Plan your data flow** from collection to analysis
4. **Organize your project structure** (folders, files)
5. **Enter data promptly** after collection
6. **Save in multiple formats** (Excel and CSV)
7. **Ensure tidy data principles** from the start

### Note

See Hadley Wickham's Tidy Data principles for best practices



## Project Design: Step 2

Create a **Metadata Sheet** that includes:

- Variable descriptions
- Units of measurement
- Collection methods

- Instrument details
- Dates and locations
- Any other relevant contextual information

| country | year | cases  | population |
|---------|------|--------|------------|
| iran    | 1999 | 143    | 150074     |
| iran    | 2000 | 200    | 200000     |
| iran    | 1999 | 87701  | 1720000    |
| iran    | 2000 | 88400  | 1740000    |
| iran    | 1999 | 212200 | 1272010    |
| iran    | 2000 | 210700 | 1280400    |

observations

## Practice Exercise 1: Pine Data Organization

### 💡 Practice Exercise 1: Pine Data Organization

Let's examine our pine needle data:

- - What naming conventions did you choose?
- - How did you organize the data?
- - How can you verify data formats (numeric vs categorical)?
- - What's your plan for organizing outputs and figures?

```
# Code to read and examine data
library(tidyverse)
library(patchwork)
library(flextable)

pine_df <- read_csv("data/pine_needles.csv")
pine_df
```

```
# A tibble: 48 × 6
  date      group      n_s  wind  tree_no length_mm
<chr>   <chr>    <chr> <chr>   <dbl>    <dbl>
1 3/20/25 cephalopods n    lee      1        20
2 3/20/25 cephalopods n    lee      1        21
3 3/20/25 cephalopods n    lee      1        23
4 3/20/25 cephalopods n    lee      1        25
5 3/20/25 cephalopods n    lee      1        21
6 3/20/25 cephalopods n    lee      1        16
7 3/20/25 cephalopods s    wind     1        15
8 3/20/25 cephalopods s    wind     1        16
9 3/20/25 cephalopods s    wind     1        14
10 3/20/25 cephalopods s    wind     1        17
# i 38 more rows
```

## Lecture 2: Data Management: Step 3

### Storage and Backup Strategy:

1. Store raw data and metadata securely
  - Save in both Excel and CSV formats
  - Consider write-protecting raw data files
2. Implement the 3-2-1 backup rule:
  - 3 total copies of data
  - 2 different storage media
  - 1 offsite location (cloud storage)
3. Establish a regular backup schedule



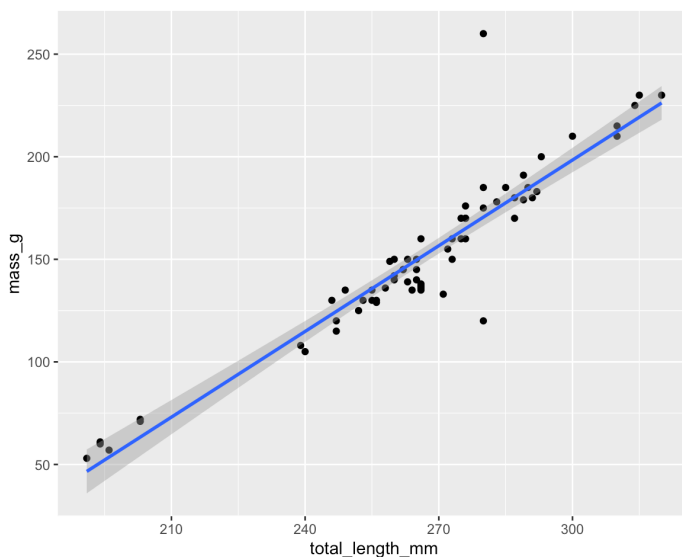
## Lecture 2: Data Management: Step 4

### Initial Data Inspection:

1. Examine data in the Environment tab
2. Run `summary()` and `glimpse()` functions
3. Create exploratory visualizations
4. Check for outliers, errors, and missing data

```
# Specify how to handle missing values during import
pine_df <- read_csv("data/pine_needles.csv",
                    na = c("", "NA", "N/A", "missing", "null"))

# Get a quick summary
summary(pine_df)
```



## Practice Exercise 2: Try plotting a histogram

## 💡 Practice Exercise 2: Try plotting a histogram of your data

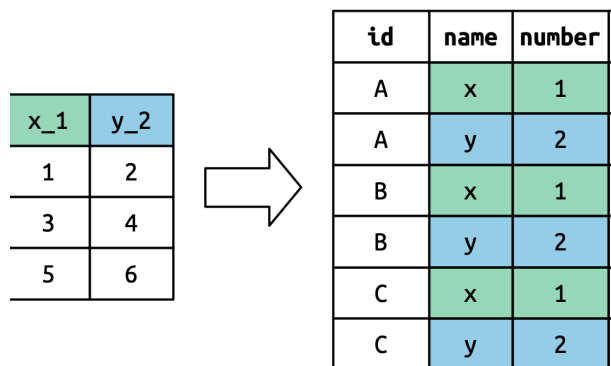
Create a histogram of pine needle lengths to check the distribution:

```
# Write your code here to make a plot
# How do you examine the data - what are the ways you think and lets try it!
```

## Lecture 2: Data Management: Step 5

### Data Cleaning:

1. Correct errors and inconsistencies
2. Replace missing values with proper NA codes
3. Document all changes made to raw data
4. Save a clean, master version (consider making read-only)
5. Keep notes on data cleaning procedures



## Lecture 2: Data Management: Step 6

### Analysis and Visualization Workflow:

1. Create exploratory visualizations
2. Summarize and transform data as needed
3. Document all analysis steps
4. Save outputs systematically

**A useful way to organize script files is number them in the order they get run.**

```
01 rewater_import_clean.R
02 rewater_initial_raw_means_figures.R
03 rewater_means_mixed_anova.R
04 rewater_anova_mixed_graphs_with_a_fun
05 rewater_anova_mixed_graphs.R
06 rewater_graphs_functions.R
07 rewater_figs_jeq_formatting.R
08 summary_statistics.R
```

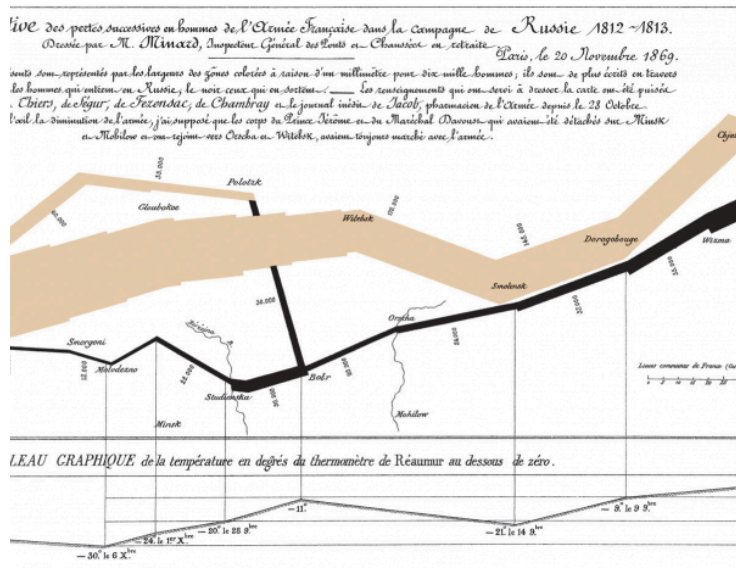
## Lecture 2: Effective Data Visualization

## Why make plots?

### Get in a group and discuss

- What is the purpose of a data visualization?
- What elements are essential in an effective plot?
- What characteristics define a “good” plot?
- What common mistakes make plots ineffective?

Napoleon’s Disastrous Invasion of Russia Detailed in an 1869 Data Visualization: It’s Been Called “the Best Statistical Graphic Ever Drawn”



## Lecture 2: Tables vs. Visualizations

### How readable are tables?

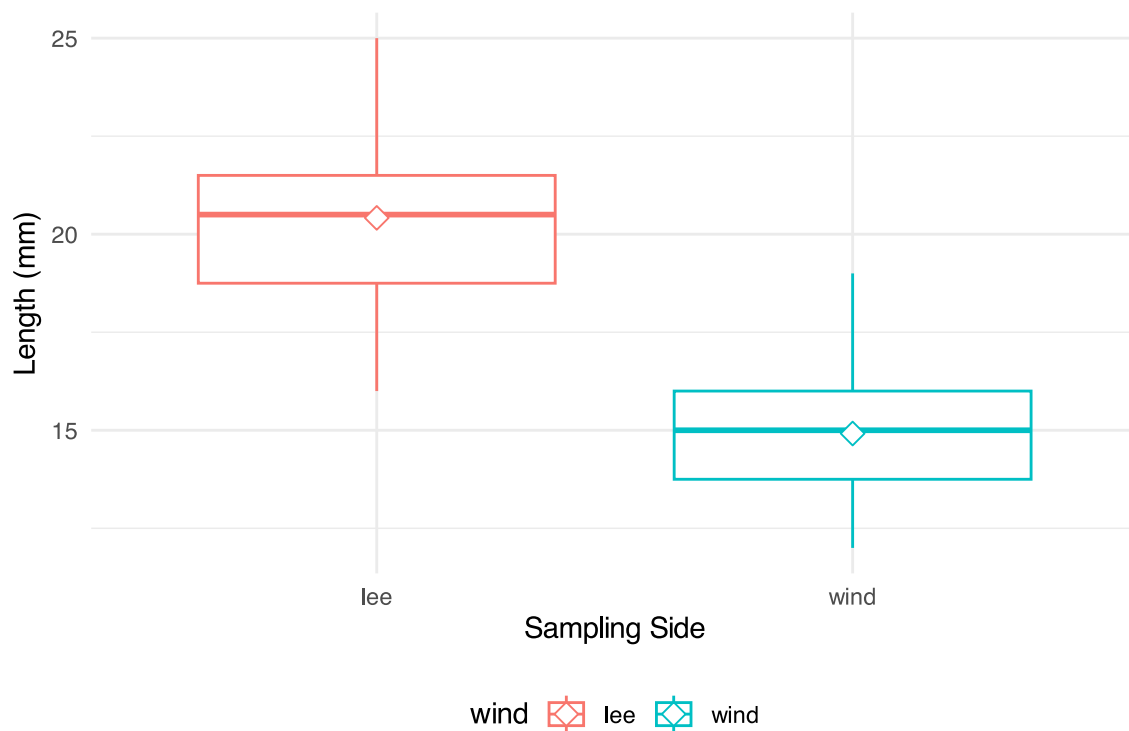
We will get to what these number mean and how to make them in the next lecture.

- Tables
  - are they useful in a presentation?

| wind | n  | mean_mm | sd_mm | min_mms | max_mm |
|------|----|---------|-------|---------|--------|
| lee  | 24 | 20      | 2.45  | 16      | 25     |
| wind | 24 | 15      | 1.91  | 12      | 19     |

## Lecture 2: Displaying data

- how does a table compare to a plot?
- Does this help?
- What is this plot?
  - if you don't explain does the audience know?



## Lecture 2: Principles of Effective Graphics

According to Tufte (2001), good scientific graphics:

1. **Show the data** without distortion
2. **Maximize data-ink ratio** (minimize non-data elements)
3. **Make large datasets coherent** and understandable
4. **Encourage comparison** between elements
5. **Reveal multiple layers** of information
6. **Serve a clear purpose** in telling your story
7. **Integrate with statistical methods** appropriately

```
# A tibble: 4 × 7
  group      mean_mm sd_mm      n se_mm conf_low conf_high
<chr>      <dbl> <dbl> <int> <dbl>   <dbl>   <dbl>
1 cephalopods  18  3.86    12  1.11    15.5    20.5
2 crayfish    18  3.86    12  1.11    15.5    20.5
3 salmon     16.3  3.94    12  1.14    13.8    18.8
4 snail       18.3  2.27    12  0.655   16.9    19.8
```

## Lecture 2: Creating Effective Graphics

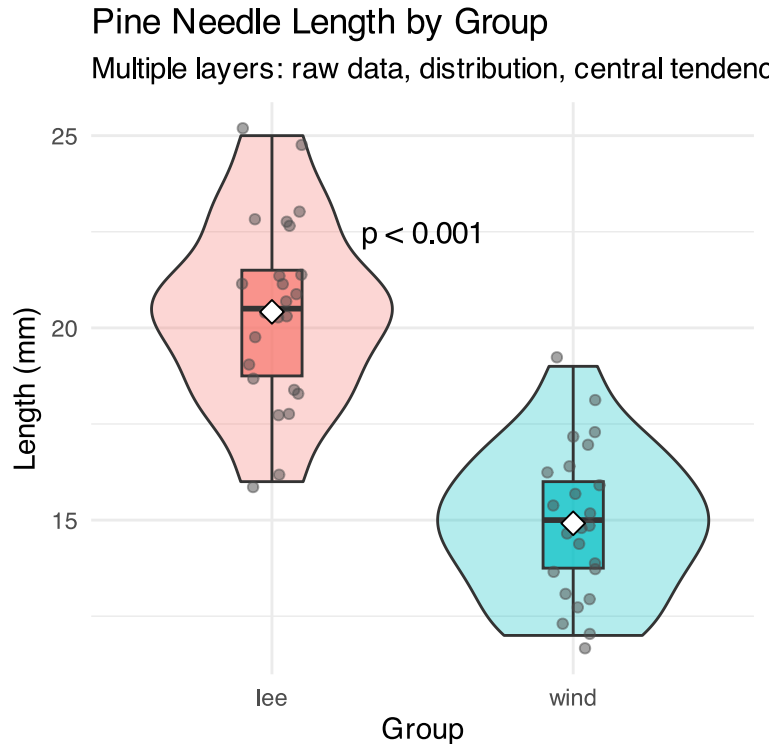
According to Tufte (2001), good scientific graphics:

- To implement these principles:
  - Focus on the data, not decorative elements
  - Ensure proportional representation of numbers
  - Provide clear and informative labels
  - Remove unnecessary elements (“chart junk”)
  - Revise and refine visualizations iteratively

## Lecture 2: Displaying data - Good Graphics

To make good graphics:

- Above all, focus on data
- Do not distort data
- Graphical representation of numbers → directly proportional to numbers
- Strive for clarity through labeling
- Maximize data-ink ratio
  - Remove non-data ink
  - Reduce redundant data ink
- Revise and redraw



## Lecture 2: Displaying data - Poor Example

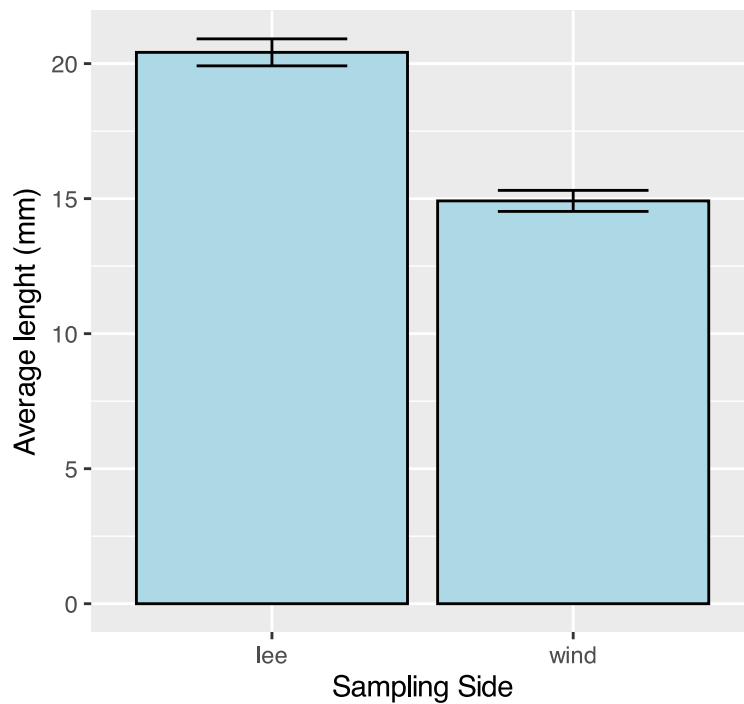
What do you think?

Does this -

- Focus on data
- Distort data
- Is it directly proportional to numbers
- Is labeling clear
- Maximize data-ink ratio
  - Remove non-data ink
  - Reduce redundant data ink
- Revise and redraw



This plot has a low data-ink ratio



## Lecture 2: Displaying data - Better Example

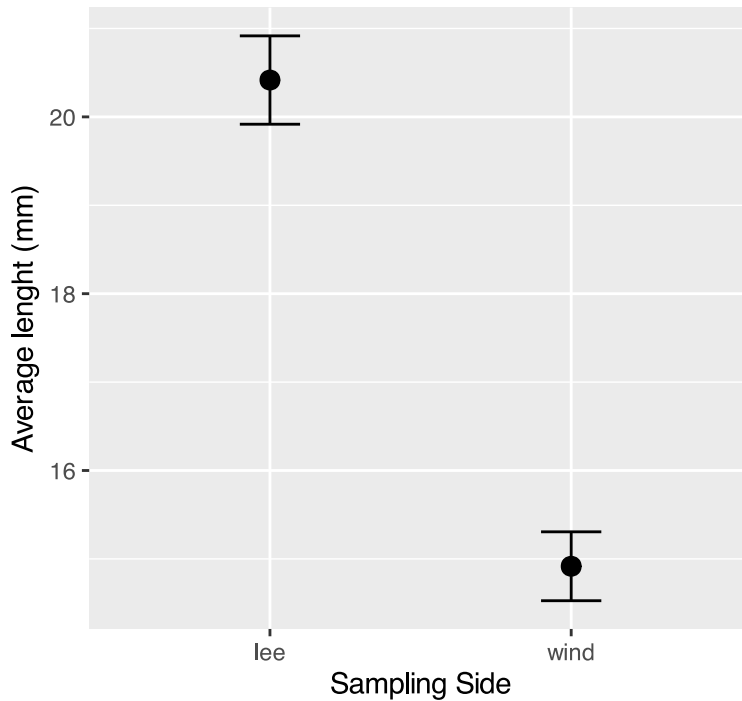
What do you think?

Does this -

- Focus on data
- Distort data
- Is it directly proportional to numbers
- Is labeling clear
- Maximize data-ink ratio
  - Remove non-data ink
  - Reduce redundant data ink
- Revise and redraw

What is one of the most common plots you make all the time?

This plot has a low data-ink ratio



## Lecture 2: Displaying data - Common Problems

### Common Visualization Problems

#### 1. Data distortion:

- Non-zero baselines on bar charts
- 3D effects that skew perspective
- Inappropriate scales

#### 2. Excessive “chart junk”:

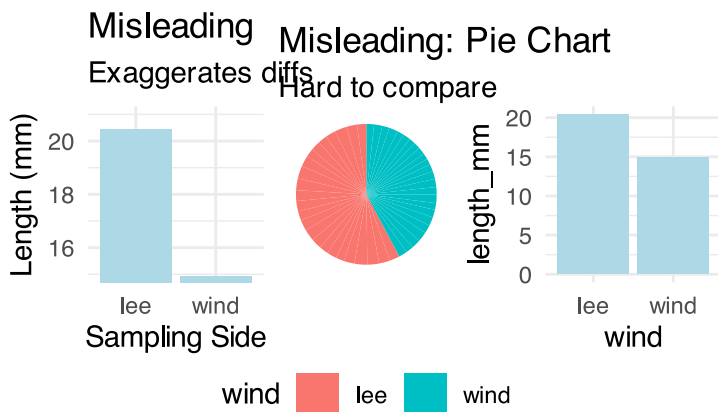
- Too many grid lines
- Unnecessary decorative elements
- Redundant information

#### 3. Poor color choices:

- Too many colors
- Non-colorblind-friendly palettes
- Colors that don't print well in grayscale

#### 4. Misleading representations:

- Pie charts with too many categories
- Dual y-axes with different scales
- Truncated axes without clear indication



## Practice Exercise 3: Basic Plots with Pine Data

💡 Practice Exercise 3: Lets try some plots with pine data first

Lets try to make some basic plots

```
# Write your code here to make a dot plot or X y plot
# How do you examine the data - what are the ways you think and lets try it!
# what is missing - hwo do you tell the effect of wind?
```

## Practice Exercise 4: Colors, Shapes, and Fills

💡 Practice Exercise 4: OK we are closer but what about colors or shape or fills

Lets try to make some more basic plots

This is free time - we will free code this....

Below are some examples of code you will need for the future

```
# Write your code here to make a dot plot or X y plot
# How do you examine the data - what are the ways you think and lets try it!
# what is missing - hwo do you tell the effect of wind?
```

## Lecture 2: Introduction to the Grammar of Graphics - ggPLOT

We will learn the anatomy of a GGplot is layers

- ggplot2 uses a **layered grammar of graphics** approach:
  1. **Data:** The dataset you're visualizing

2. **Aesthetics:** Mapping variables to visual properties
  3. **Geometries:** The visual elements representing data
  4. **Facets:** Splitting visualization into subplots
  5. **Statistics:** Statistical transformations of the data
  6. **Coordinates:** The space in which data is plotted
  7. **Themes:** Overall visual style of the plot
- We have aesthetics

## Lecture 2: Building a ggplot Visualization

### Key Components:

1. **Aesthetics (aes)** map variables to visual properties:
  - x and y positions
  - color, fill, shape, size, alpha
  - group, linetype
2. **Geometries (geom\_\*)** determine how data is displayed:
  - `geom_point()`: Scatter plots
  - `geom_line()`: Line graphs
  - `geom_boxplot()`: Box-and-whisker plots
  - `geom_violin()`: Violin plots
  - `geom_histogram()`: Histograms
  - `geom_bar()`: Bar charts
3. **Position adjustments** control how elements are arranged:
  - `position_dodge()`: Side-by-side elements
  - `position_jitter()`: Add random noise to points
  - `position_stack()`: Stack elements on top of each other
4. **Labels and annotations** provide context:
  - `labs()`: Title, subtitle, caption, axis labels
  - `annotate()`: Add text, shapes, etc.

## Lecture 2: Fine-tuning your visualizations

1. **Colors, fills, and shapes:**

```
scale_color_manual(
  values = c("wind" = "darkblue", "lee" = "darkred"),
  labels = c("wind" = "Windward", "lee" = "Leeward")
)
```

2. **Coordinate systems:**

```
coord_cartesian(ylim = c(10, 30)) # Zoom in without dropping data
```

3. **Themes:**

```
theme_minimal() +
theme(
  axis.title = element_text(size = 14),
  legend.position = "bottom"
)
```

4. **Combining plots with patchwork:**

```
plot1 + plot2 + plot_layout(ncol = 2)
```

## Practice Exercise 5: Publication-Quality Plot

### 💡 Practice Exercise 4: Creating a Publication-Quality Plot

Create a fully customized plot that would be suitable for publication:

```
# Create a publication-quality plot
pine_df %>%
  ggplot(aes(x = wind, y = length_mm, fill = wind)) +
  geom_violin(alpha = 0.4) +
  geom_boxplot(width = 0.2, alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.1, alpha = 0.5, color = "gray30", size = 2) +
  stat_summary(fun = mean, geom = "point", shape = 23, size = 3, fill = "white") +
  labs(
    title = "Pine Needle Length Varies with Wind Exposure",
    subtitle = "Needles on the leeward side tend to be longer",
    x = "Tree Side",
    y = "Needle Length (mm)",
    caption = "Data collected Spring 2023") +
  scale_fill_manual(
    values = c("wind" = "#1b9e77", "lee" = "#d95f02"),
    labels = c("wind" = "Windward", "lee" = "Leeward")) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12, color = "gray30"),
    axis.title = element_text(face = "bold"),
    legend.title = element_blank(),
    legend.position = "bottom")
```

## Key Takeaways

1. **Plan your data management** from the beginning
  - Consistent naming conventions
  - Good organization
  - Regular backups
2. **Make your data tidy** from the start
  - One observation per row
  - One variable per column
  - One value per cell
3. **Create effective visualizations** by:
  - Focusing on data, not decoration
  - Using appropriate plot types
  - Following good design principles
  - Customizing for clear communication
4. **Master the grammar of graphics** to:
  - Build plots layer by layer
  - Communicate patterns clearly
  - Tell compelling stories with data

## Next Steps

- Practice creating different types of plots
- Learn to combine multiple plots effectively
- Explore statistical transformations in ggplot2
- Develop a consistent visualization style